
HTAST-ALIGN: A TWO-STAGE SOUND IDENTIFICATION MODEL

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ABSTRACT

In the keyword recognition subtask of audio classification, conventional Transformer models typically convert the audio classification task into an image recognition problem, treating the target keywords as the classification labels. By decomposing the keyword recognition task into two stagesphoneme recognition followed by keyword recognitionthis paper achieves an accuracy of 0.9854 ± 0.0001 on the Google Speech Commands v2 dataset. See all of my code at <https://github.com/FragrantOrchid/HTSAT-Align>

Keywords Audio classification · Transformer

1 Introduction

We observe that when deploying audio classification models such as HTSAT[Chen et al., 2022] for keyword recognition, systems typically employ direct keyword-level labels as classification targets. However, performance degrades significantly on highly similar keywords, particularly when the phonetic sequence of one term constitutes a substring of another. To address this limitation, we treat the outputs of the Transformer module within the HTSAT architecture as temporal phoneme-level features, which are subsequently processed by a Bidirectional LSTM (Bi-LSTM)[Sak et al., 2014] module to map them into keyword classification predictions. Notably, the Transformer backbone can be pre-trained on a large-scale speech corpus before undergoing task-specific fine-tuning on the target dataset. Experimental results on the Google Speech Commands v2[Warden, 2018] dataset demonstrate that our method achieves an accuracy of 0.9854.

2 Model Architecture

2.1 GaussianSpecAugment

Traditional SpecAugment[Park et al., 2019] typically relies on the combined application of time and frequency masking. However, in keyword recognition tasks, human speech exhibits significantly higher information density than general acoustic sounds, rendering the conventional block-wise masking prone to obscuring critical phonetic or contextual features. Drawing inspiration from SSAST[Gong et al., 2022], this paper introduces a Gaussian-smoothed masking scheme wherein discrete mask patches spontaneously aggregate into coherent clusters under configurable hyperparameters. This adaptive approach effectively preserves essential acoustic-linguistic information even under high masking ratios.As Figure 1 shows.

2.1.1 SwinTransformer

We adopt the same Swin Transformer architecture as HTSAT. The key distinction lies in the deliberate control of input and output temporal resolutions, such that the output tensor of the Swin Transformer is directly aligned with strong-label classification targets at the temporal resolution level, while the classification head consists solely of a linear layer. Through this design, the model is compelled to delegate the vast majority of its classification capacity to the Swin Transformer module, thereby maximally preserving pre-trained knowledge during downstream data transfer.

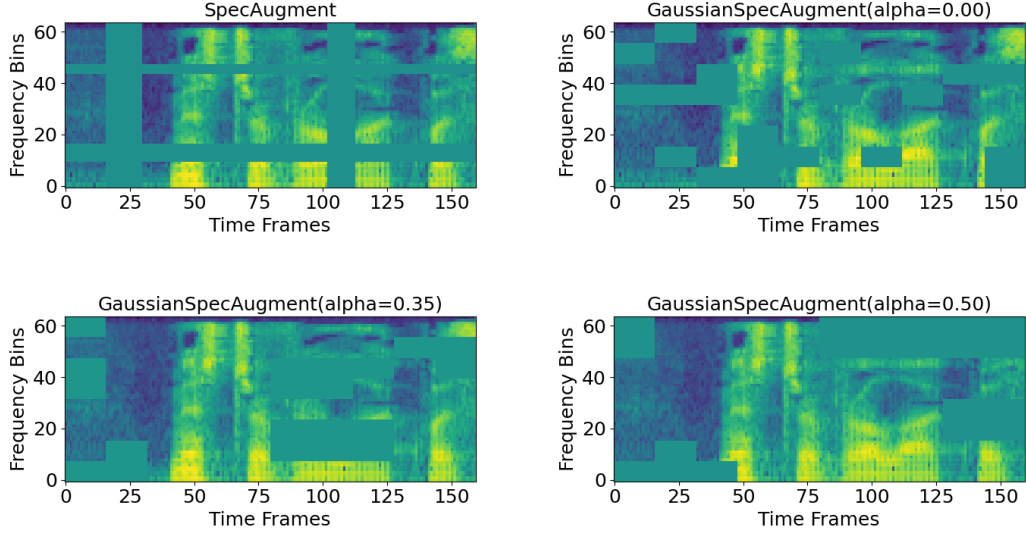


Figure 1: different-spec

2.2 Bi-LSTM

A Bidirectional LSTM (Bi-LSTM) is employed to transform the sequence of phoneme-level temporal features into concrete keyword classification labels. Bi-LSTM architectures excel at distilling information from sequences and are inherently sensitive to the temporal dynamics of input features. In our implementation, we adopt a two-layer Bi-LSTM module wherein no intermediate projections are applied within the recurrent layers; instead, only the final output is projected onto the keyword classification labels in a single step.

3 Train Method

3.1 PreTrain

During the pre-training stage, the Swin Transformer module is trained on the LibriSpeech-Alignments [Gilkey and Lugosch, 2023] dataset using strong-label data, enabling the direct extraction of phoneme-level information at individual time frames. Furthermore, by leveraging the contextual modeling capacity of the Swin Transformer, the model is also capable of capturing stress patterns of vowel phonemes and the positional information of vowel syllables within words. The detailed configuration of classification labels is presented in Table 3.1.

Table 1: Labels used in pretrain

Index	Name	Meaning
0-38	[IPA]	IPA
39	spn	Not Speech
40	0	Unstressed
41	1	Primary stress
42	2	Secondary stress
43	Begin	Vowel in initial syllable
44	Inside	Vowel in medial syllable
45	End	Vowel in final syllable
46	Single	Vowel in only syllable

3.2 Train

For training on the Google Speech Commands v2 dataset, we adopt a two-stage strategy. In the first stage, the parameters of the Swin Transformer module are frozen, and only the Bi-LSTM module is trained for a small number of epochs. In the second stage, all parameters are unfrozen, the learning rate is reduced, and the entire model is fine-tuned end-to-end.

4 Result

Table 2: Test on Speech Command V2

Model	Pretrain	Accuracy(%)
HTSAT-Align	✓	98.54
EAT[Chen et al., 2024]	✓	98.3 $\pm$ 0.04
AST-S[Gong et al., 2021]	×	98.11 $\pm$ 0.05
PDC[Chrysos et al., 2022]	×	97.8
ASM-RH[Ji et al., 2024]	×	96.51
EfficientLEAF[Schlüter and Gutenbrunner, 2022]	×	95.2
LEAF[Schlüter and Gutenbrunner, 2022]	×	95.1
melspect[Schlüter and Gutenbrunner, 2022]	×	95.1

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